# Advanced Animations & Gestures in Flutter

Topic 08 — Cross-Platform Development

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Flutter Motion | Gesture Arena | 60/120 FPS VSync

# Agenda

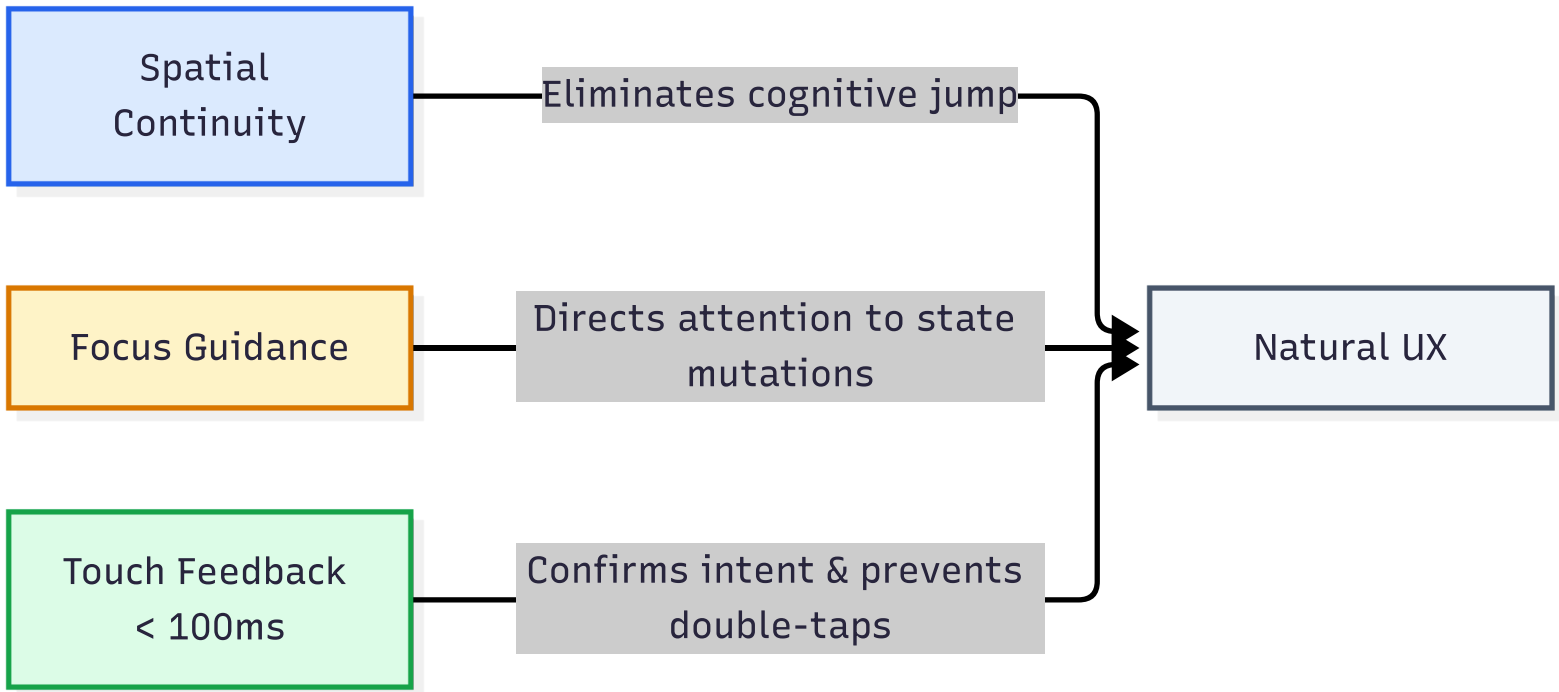
## Part 1: Core Mechanics

1. Animation & UX Roles
2. Core Terminology & Concepts
3. Technical Requirements
4. Flutter Motion Architecture
5. Implicit vs Explicit Matrix
6. Explicit Lifecycle & Memory Safety
7. Gesture Arena & Disambiguation
8. UX Latency Thresholds

## Part 2: Implementation & Evidence

9. Architecture & Motion Tokens
10. Implicit Motion: State Interpolation
11. Explicit Gestures: Spring Physics
12. Automated Test Suite
13. Decision Matrix
14. References

# 1. Animation & UX Roles



# 2. Core Terminology & Concepts

### Implicit Animations

**Declarative (Tự nội suy):** Chỉ khai báo giá trị đích; framework tự tính toán vi phân qua từng khung hình.

### Explicit Animations

**Imperative (Chủ động điều khiển):** Dùng `AnimationController` để phát, lặp, đảo chiều và bám cử chỉ tay.

### Gesture Arena

**Đấu trường phân giải:** Cơ chế trọng tài giải quyết xung đột khi nhiều cử chỉ cùng bắt một điểm chạm.

### Non-blocking UX

**Không khóa tương tác:** Chuyển động chạy độc lập, không đóng băng luồng nhận sự kiện cảm ứng.

### VSync (Vertical Synchronization)

**Đồng bộ làm tươi phần cứng:** Tín hiệu xung nhịp từ màn hình (60Hz / 120Hz). Flutter dùng `Ticker` khóa chặt tốc độ tính frame theo nhịp VSync, giúp triệt tiêu hiện tượng xé hình (screen tearing) và tiết kiệm pin khi màn hình nghỉ.

# 3. Technical Requirements

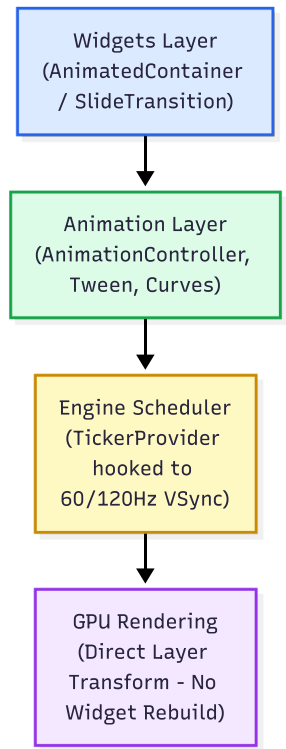
## Theoretical Foundations

- **Implicit Animations:** Declarative target interpolation.
- **Explicit Animations:** Imperative `AnimationController`.
- **Gesture Arena:** Touch conflict disambiguation.
- **Non-blocking UX:** Unconstrained touch pipeline.

## Practical Deliverables

- **Implicit Card:** Shape, color, and size transitions.
- **Draggable Spring Card:** 2D pan + spring physics.
- **Morphing Button:** Non-blocking async feedback.
- **Automated Verification:** 5/5 test pass, 0 leaks.

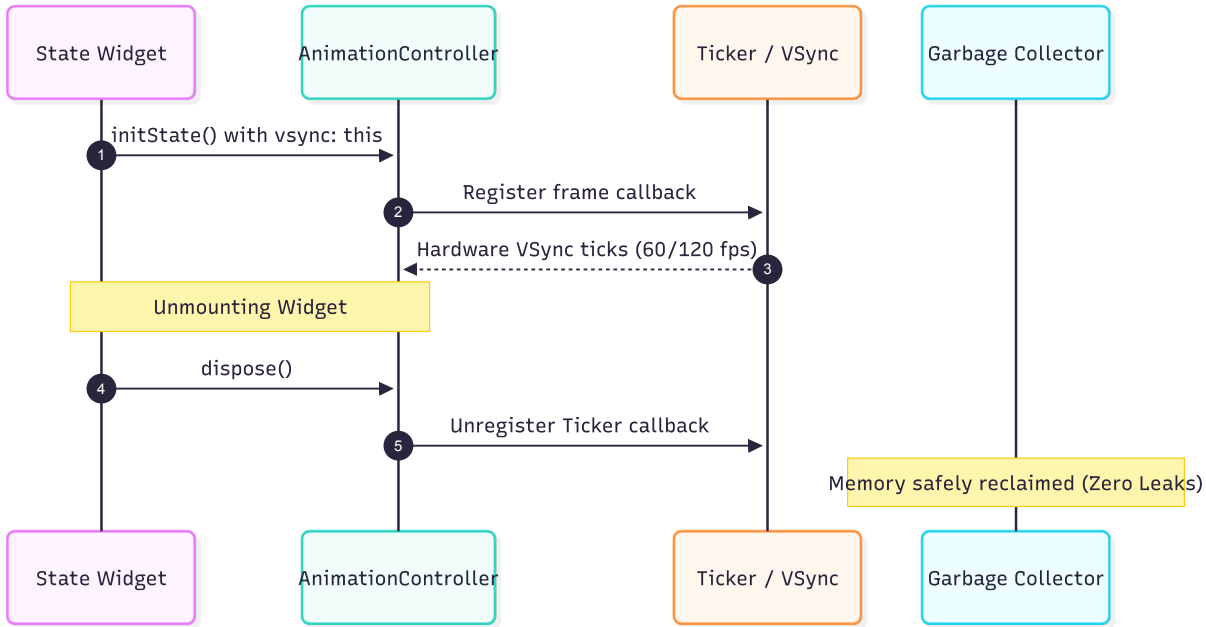
# 3. Flutter Motion Architecture



# 4. Implicit vs Explicit Matrix

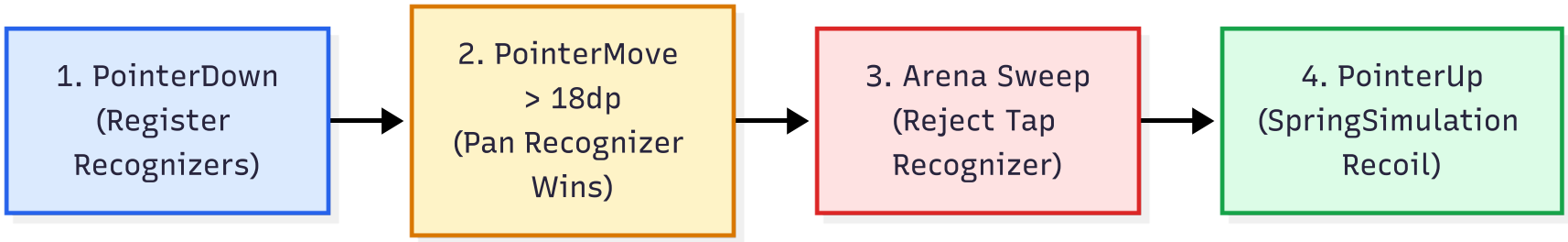
| Dimension | Implicit Animations | Explicit Animations |
| --- | --- | --- |
| **Programming Model** | Declarative | Imperative |
| **Controller Management** | Managed internally by Framework | Manual `AnimationController` |
| **Memory Leak Risk** | **0% (Zero Leak)** | **High** (If unreleased in `dispose()`) |
| **Repeat / Reverse** | Not supported | Full control (`repeat()`, `reverse()`) |
| **Gesture Binding** | Incompatible | Direct pan coordinate sync |

# 5. Explicit Lifecycle & Memory Safety




**Memory Leak Hazard:** Omitting `dispose()` keeps `TickerProvider` retained in the Engine Scheduler via a **Strong Reference**, preventing GC unmount.

# 6. Gesture Arena & Disambiguation




- **2D Pan Tracking:** Real-time translation with dynamic tilt angle calculations.
- **SpringSimulation:** Natural damped harmonic recoil driven by release velocity.

# 7. UX Latency Thresholds

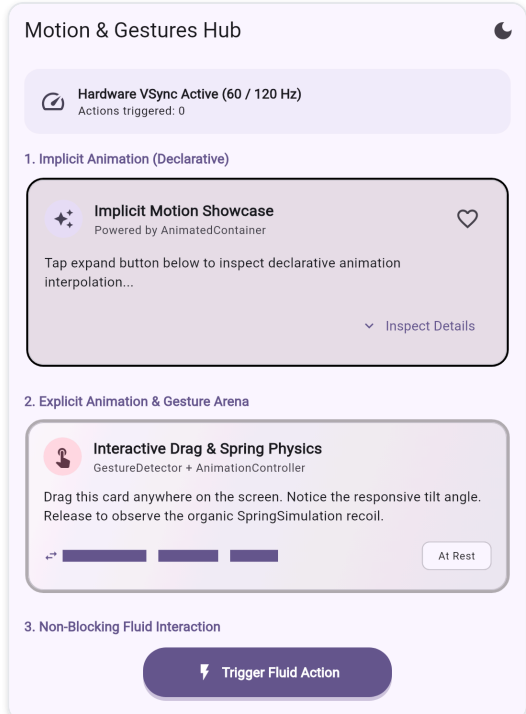
| Latency | Classification | Practical Application |
| --- | --- | --- |
| **< 100 ms** | Instant | Touch ripple, physical pressed states |
| **150 – 350 ms** | Natural (Gold Standard) | Navigation, element expansion, modal transitions |
| **> 500 ms** | Sluggish | Perceived as system lag; causes user friction |

- **Non-Blocking UX:** Keep animations interruptible; never freeze user gestures.

# 8. Architecture & Motion Tokens

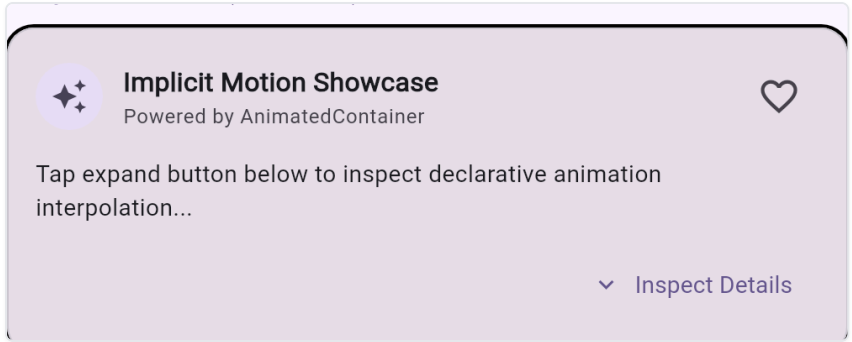
Demo App Structure ( `week_04/app` ):

- **MotionTokens (8dp Grid):**
  `space8` , `space16` , `space24` .
- **Three Independent Subsystems:**
  1. `ImplicitMotionCard` : Declarative interpolation.
  2. `DragSpringCard` : Gesture Arena + Spring physics.
  3. `MorphingActionButton` : Non-blocking async feedback.
- **VSync Synchronization:**
  Locked to 60Hz / 120Hz display refresh.

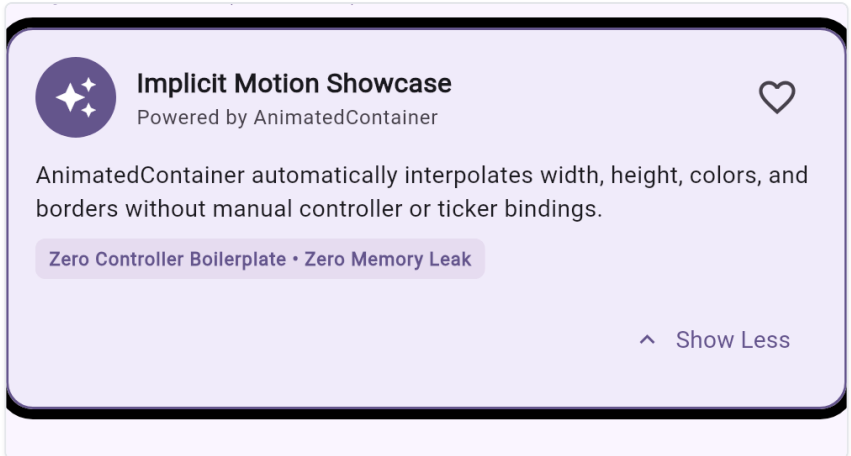


# 9. Implicit Motion: State Interpolation

Rest State (At Rest)

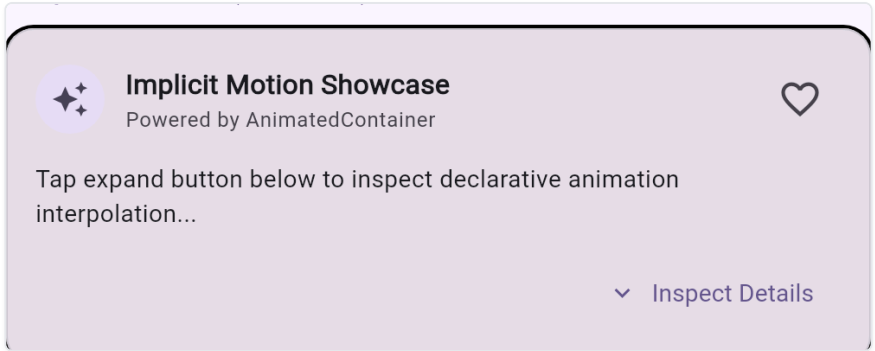



Expanded State (Interpolated)



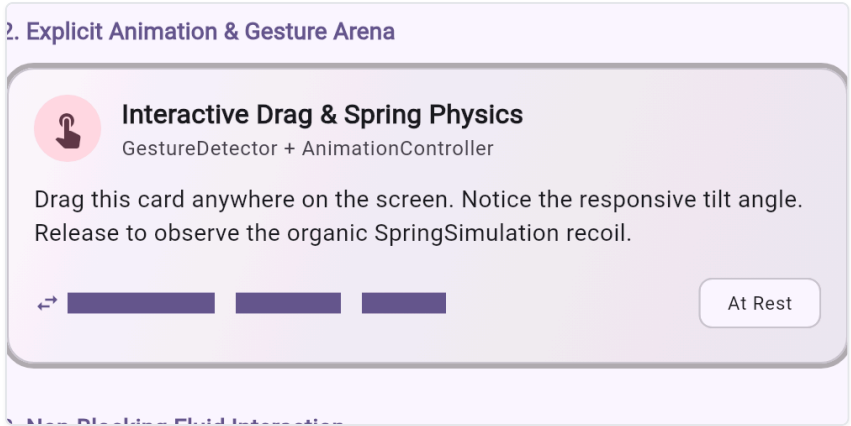
# 9. Implicit Motion: Mechanics

- **AnimatedContainer:** Interpolates property deltas across duration and curve.
- **Automatic State Sync:** Recomputes layout without external controllers.
- **Zero Resource Leaks:** Framework guarantees lifecycle teardown.

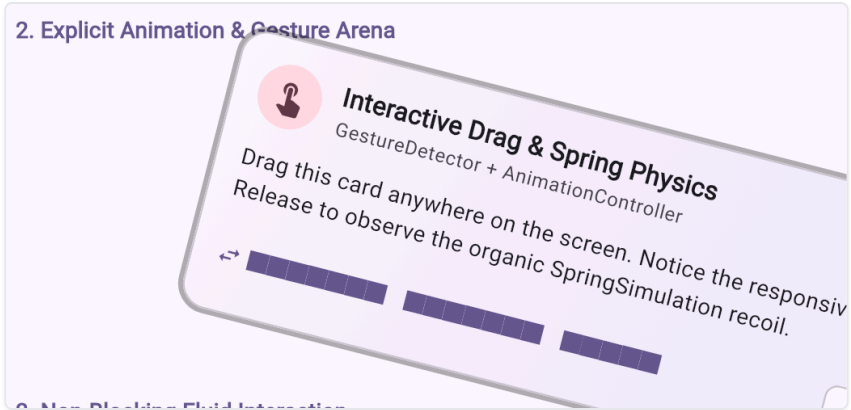


# 10. Explicit Gestures: Spring Physics

Rest State (Offset 0, 0)

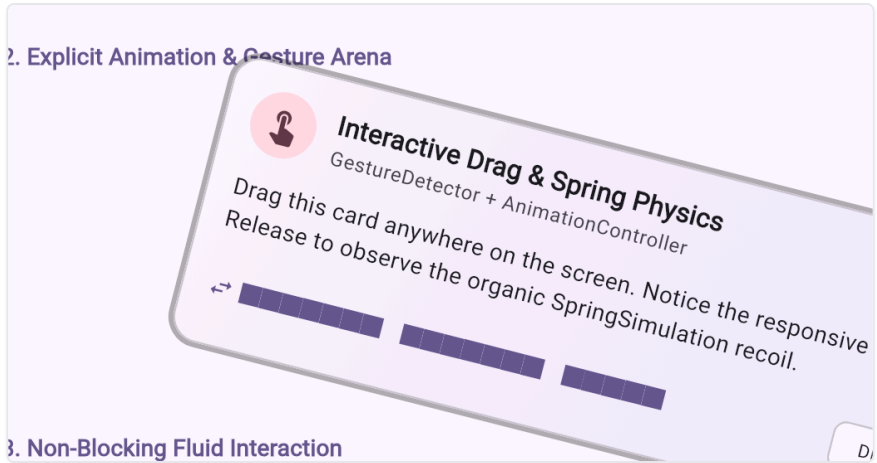



Dynamic Tilt State (Under Drag)



# 10. Explicit Gestures: Mechanics

- **GestureDetector (`onPanUpdate`):** Synchronizes card translation and rotational tilt with pointer coordinates.
- **SpringSimulation (`onPanEnd`):** Runs physical recoil back to `(0, 0)` preserving release inertia.
- **Resource Teardown:** Explicit disposal in `State.dispose()`.

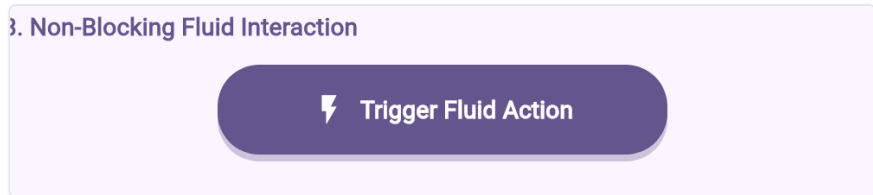


# 11. Automated Test Suite

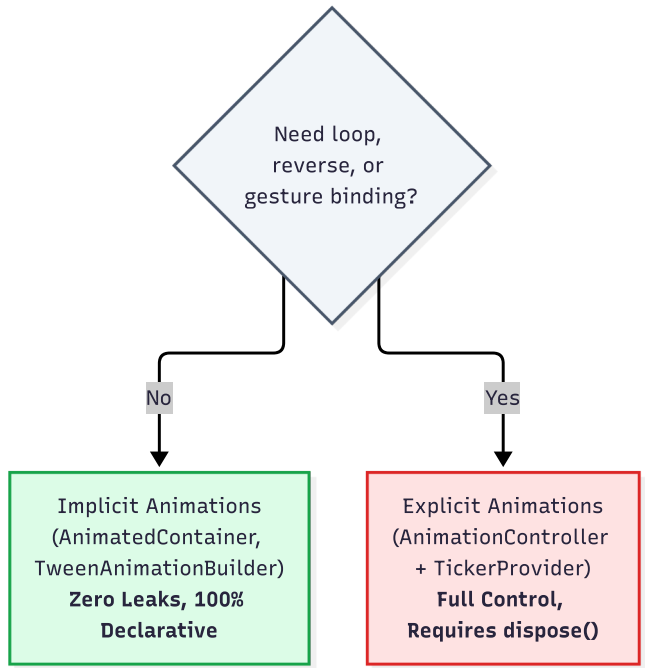
Verification Scenarios (5 Test Specs):

- **Widget Rendering:** Validates Material 3 layout structure.
- **Interpolation:** Verifies smooth frame value progression.
- **Gesture Recoil:** Asserts 2D pan tracking and spring return.
- **Non-blocking Async:** Verifies UI pipeline during background tasks.
- **Resource Teardown:** Asserts 100% ticker disposal on unmount.

Morphing Button Async State



# 12. Decision Matrix



# 13. References

- **Flutter Animations Overview:** docs.flutter.dev/ui/animations/overview
- **Implicit Animations Guide:** docs.flutter.dev/ui/animations/implicit-animations
- **Gestures & Disambiguation Arena:** docs.flutter.dev/ui/interactivity/gestures
- **AnimationController Class:** api.flutter.dev/flutter/animation/AnimationController-class.html
- **GestureArenaManager Class:** api.flutter.dev/flutter/gestures/GestureArenaManager-class.html
- **Material Design 3 Motion:** m3.material.io/styles/motion/overview/how-it-works